

1 **(a)** Use the List of formulae (MF19) to find $\sum_{r=1}^n (2r+1)$ in terms of n , simplifying your answer. [2]

[illegible]

(b) Show that $\frac{2r+1}{(r^2+1)(r^2+2r+2)} = \frac{1}{r^2+1} - \frac{1}{r^2+2r+2}$. [1]

[illegible]

- (c) Use the method of differences to find $\sum_{r=1}^n \frac{2r+1}{(r^2+1)(r^2+2r+2)}$. [2]

[illegible]

- (d) Deduce the value of $\sum_{r=1}^{\infty} \frac{2r+1}{(r^2+1)(r^2+2r+2)}$. [1]